

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2013-14

Paper No. CHB-502: Inorganic Chemistry-III

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer five questions in total, including Question 1 which is compulsory.***Question 1.** Answer all parts of the following:

[5 × 2 = 10]

- (a) Why do 3d-elements form stable complexes with nitrogen, oxygen, and fluorine donor ligands, whereas 4d and 5d elements prefer phosphorus, sulfur, and chlorine donors?
- (b) What is the chelate effect? Give one example.
- (c) Why is $[\text{Co}(\text{NH}_3)_6]^{2+}$ oxidized easily to $[\text{Co}(\text{NH}_3)_6]^{3+}$?
- (d) Why do 3d metals form both low-spin and high-spin complexes, whereas 4d and 5d metals form only low-spin complexes?
- (e) Draw the structure of Zeise's salt.

Question 2.

- (a) For the formation of $[\text{Fe}(\text{bipy})_3]^{2+}$ from $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ and bipyridyl, the stepwise formation constants lie in the order $K_1 > K_2 < K_3$. Explain this anomaly. [6]
- (b) Define and discuss the Jahn-Teller effect. Predict which of the following octahedral complexes show Jahn-Teller distortion: $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$, $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$, and $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$. [8]

Question 3.

- (a) Discuss the Crystal Field Theory. Describe the splitting of d-orbitals in octahedral, tetrahedral, and square-planar complexes. What is the relation between octahedral and tetrahedral splitting energy ($\Delta_t = \frac{4}{9}\Delta_o$)? [8]
- (b) Comment on the hydration energies of divalent transition metal ions (M^{2+}) of the 3d series. [6]

Question 4.

- (a) The reaction between $[\text{PtCl}(\text{dien})]^+$ and I^- is first-order in each reactant. Discuss the mechanism of substitution in square-planar complexes, highlighting the pathway and intermediate involved. [8]

- (b) Discuss the factors responsible for the catalytic activity of Wilkinson's catalyst. Describe the mechanism of the hydrogenation of olefins using this catalyst. [6]
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Question 5.

- (a) What is haptacity? Show that the haptacity of a ligand varies from one organometallic compound to another using cyclopentadienyl and allyl complexes. [8]
- (b) Describe the synthesis and mechanism of action of the Ziegler-Natta catalyst for the stereospecific polymerization of propylene. [6]